

Bases and Dimension

Bases

Defn. A **basis** B of a subspace is a set such that (i) B is linearly independent, and (ii) the span of B is the whole subspace.

Bases of $\mathbb{R}^2$

The standard basis of $\mathbb{R}^2$ is $\{ (1, 0), (0, 1) \}$.

But any two vectors that are linearly independent (that is, not parallel) form a basis.

However, any trio of vectors is linearly dependent, while no one vector spans the space.

Bases and Matrices

Fact. *An $n \times n$ matrix is invertible if and only if its columns form a basis of $\mathbb{R}^n$.*

Bases and Representation

Fact. *If B is a basis for subspace S , then every element in S is **uniquely** expressible as a linear combination of vectors of B*

For example, in $\mathbb{R}^2$, one can uniquely identify a point given any two linearly independent vectors u and v : go so many units in the direction of u and then so many units in the direction of v .

Dimension

Fact. *All bases of a subspace have the same number of elements.*

Defn. *The **dimension** of a subspace is the number of elements in a basis.*

Examples of Dimension

- ⤿ $\mathbb{R}^d$ has dimension d .
- ⤿ A line has dimension 1.
- ⤿ A plane has dimension 2.
- ⤿ The space $\{0\}$ has dimension 0 and an empty basis.

Building Bases Knowing Dimension

Fact. *If subspace S has dimension p , then*

- (a) Any linearly independent set of p vectors forms a basis of S .*
- (b) Any set of p vectors whose span is S forms a basis of S .*

That is, a basis is simultaneously a spanning set that is as small as possible, and a linearly independent set that is as large as possible.

Summary

A basis B of a subspace is a set such that B is linearly independent, and the span of B is the whole subspace. All bases of a subspace have the same number of elements called its dimension.

Every element of the subspace is uniquely expressible as a linear combination of vectors of B . An $n \times n$ matrix is invertible if and only if its columns form a basis of $\mathbb{R}^n$.